

Dharshan Paramasivan

dharshanparamasivan@gmail.com | +91 96776 54288

[LinkedIn](#) | [GitHub](#)

Coimbatore, Tamil Nadu, India

ACADEMICS CREDENTIALS

M.Sc. in Software Systems

PSG College of Arts & Science

Coimbatore, India

Percentage : 82% | 2022 – Present

HSC

Hindusthan Matriculation and Higher Secondary School

Coimbatore, India

Percentage : 84% | 2021-2022

INTERNSHIP

I. Fulcrum Technologies Inc., India

2025

Role: Intern | Tester | 3 Months

Responsibilities: Executed end-to-end test automation using Selenium WebDriver and Cucumber (BDD) in Eclipse IDE, improving test efficiency by **30%** and reducing manual testing effort. Designed and maintained reusable test scripts for web applications, ensuring consistent validation of key functionality.

Achievements: Transformed manual test cases into structured automation scripts, reducing testing time by **40%** and increasing test coverage.

II. Pinsphere Solutions Pvt. Ltd., India

2025

Role: Intern | Full Stack Developer | 3 Months

Responsibilities: Developed full-stack web applications using Django and PostgreSQL, improving backend performance by **25%** through optimized database integration and scalable architecture. Built and optimized RESTful API endpoints for efficient data handling.

Achievements: Spearheaded the development of scalable web applications, ensuring reliable backend performance and deployment-ready solutions.

SKILLS

Programming Languages: Java • Python • JavaScript • SQL

Frameworks & Technologies: MERN Stack (MongoDB, Express.js, React.js, Node.js) • Django • FastAPI • REST APIs • Next.js

Databases: PostgreSQL • MongoDB • MySQL

Testing & QA: Selenium WebDriver • Cucumber (BDD) • Apache JMeter • Postman

Cyber Security: Burp Suite • OWASP ZAP • API Security Testing

Developer Tools: Git • Docker • Eclipse IDE • Visual Studio Code • NGINX • Supabase

Operating Systems: Windows • Kali Linux

Soft Skills: Analytical Thinking • Team Collaboration • Leadership • Time Management • Adaptability • Communication

PROJECTS

I. Vulnerability Assessment and Penetration Testing on API

2024

Tech Stack: Burp Suite | Postman | OWASP ZAP

Objective: To evaluate the security of APIs by identifying vulnerabilities and testing potential attack vectors, and to ensure secure communication and proper data handling practices are in place.

Outcome: Conducted API security testing using Burp Suite, Postman, and OWASP ZAP, identifying critical vulnerabilities and improving system security with actionable fixes.

II. Appointment Management System

2025

Tech Stack: Django, Python, PostgreSQL

Objective: To develop a web-based system for efficient appointment booking and scheduling, and to implement secure access control for different user roles.

Outcome: Developed a Django-based appointment system with role-based access control, improving scheduling efficiency and optimizing database performance.

III. Automation Testing of Web Application

2025

Tech Stack: Selenium, Cucumber, Java, Eclipse IDE

Objective: To validate web application functionality through automated end-to-end testing and ensure reliability of critical workflows such as login and data entry.

Outcome: Built an end-to-end automation framework using Selenium WebDriver and Cucumber, improving test reliability and reducing manual testing effort by 30%. Automated key workflows, increasing test coverage and execution efficiency.

IV. Machinery Health Monitoring System (VortexIoT)

2026

Tech Stack: Next.js, FastAPI, PostgreSQL, Kafka, InfluxDB, AWS, Docker

Objective: To develop an Industrial IoT-based machinery health monitoring system that collects real-time sensor telemetry, detects equipment anomalies using machine learning, and enables scalable cloud-based predictive maintenance.

Outcome: Developed a real-time predictive maintenance platform using Next.js, FastAPI, Kafka, and InfluxDB with PostgreSQL for data management. Implemented ML-based anomaly detection and deployed the system on AWS using Docker, ensuring scalable, high-availability monitoring of industrial machinery.

Live Demo: <https://machinary-one.vercel.app/>

Courses and Certifications

API Penetration Testing – APIsec University

2023

Java for Android - Vanderbilt University (Coursera).

2024

Python for Data Science, AI & Development - IBM (Coursera).

2025

Introduction to Internet of Things - IIT Bombay (Coursera).

2025

Database Management System - NPTEL.

2025